

Algebra II with Trigonometry

Chapter 5.3

Olympiad Solutions

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[4401269_alg-2trig-h-chp-5-3-note-docx.pdf](#)

Solutions

1. If $\sin \theta + \cos \theta = \frac{7}{5}$, **determine without a calculator which quadrant(s) θ could lie in.**

Solution

We know that $-\sqrt{2} \leq \sin \theta + \cos \theta \leq \sqrt{2}$, since

$$\sin \theta + \cos \theta = \sqrt{2} \sin (\theta + 45^\circ)$$

Here, $\frac{7}{5} \approx 1.4 < \sqrt{2} \approx 1.4142$, so this sum is possible.

Both $\sin \theta$ and $\cos \theta$ are individually bounded by $[-1, 1]$. If their **sum is positive and large** (> 1), both must be positive and close to 1. Thus:

- $\sin \theta > 0$
- $\cos \theta > 0$

These are **simultaneously true only in Quadrant I.**

Conclusion: θ can only be in Quadrant I.

2. **Find all values of k such that the equation**

$\tan(k\pi + \frac{\pi}{6}) = \tan(\frac{5\pi}{6} - k\pi)$ **holds for integer k between -10 and 10 .**

Solution

Recall that $\tan(\theta + n\pi) = \tan \theta$ for integer n .

Let's simplify both sides:

Left: $\tan(k\pi + \frac{\pi}{6}) = \tan(\frac{\pi}{6})$, since addition of $k\pi$ does not change the value.

Right: $\tan(\frac{5\pi}{6} - k\pi) = \tan([-k]\pi + \frac{5\pi}{6})$. Again, $\tan(m\pi + \alpha) = \tan \alpha$, so:

$$\tan(\frac{5\pi}{6} - k\pi) = \tan(\frac{5\pi}{6})$$

Thus, equation reduces to:

$$\tan\left(\frac{\pi}{6}\right) = \tan\left(\frac{5\pi}{6}\right)$$

$$\text{But } \tan\left(\frac{\pi}{6}\right) = \frac{1}{\sqrt{3}}; \tan\left(\frac{5\pi}{6}\right) = -\frac{1}{\sqrt{3}}.$$

So,

$$\frac{1}{\sqrt{3}} = -\frac{1}{\sqrt{3}}$$

which is impossible.

Therefore, there is no integer k in $-10 \leq k \leq 10$ for which the equation holds.

Conclusion: No such integer k in the given range.

3. Without explicit calculation, compare the areas of two triangles: Triangle A with sides 10 and 24 and included angle 60° , and Triangle B with sides 12 and 20 and included angle 90° . Which triangle has a larger area?

Solution

Area formula:

$$\text{Area} = \frac{1}{2}ab \sin C$$

for two sides a, b and included angle C .

Triangle A:

$$\text{Area}_A = \frac{1}{2} \times 10 \times 24 \times \sin 60^\circ = 120 \times \frac{\sqrt{3}}{2} = 60\sqrt{3}$$

Triangle B:

$$\text{Area}_B = \frac{1}{2} \times 12 \times 20 \times \sin 90^\circ = 120 \times 1 = 120$$

Now compare $60\sqrt{3}$ and 120.

But $\sqrt{3} \approx 1.732 \implies 60\sqrt{3} \approx 103.9 < 120$.

Conclusion: Triangle B has a larger area.

4. Let θ be an angle in standard position with $\cos \theta = -\frac{4}{5}$ and θ in the third quadrant. Compute the exact value of $\tan 2\theta$.

Solution

Given:

- $\cos \theta = -\frac{4}{5}$
- θ in Quadrant III ($\sin \theta < 0, \cos \theta < 0$)

Find $\sin \theta$ using $\sin^2 \theta + \cos^2 \theta = 1$:

$$\sin^2 \theta = 1 - \left(-\frac{4}{5}\right)^2 = 1 - \frac{16}{25} = \frac{9}{25}$$

So $\sin \theta = -\frac{3}{5}$ (since sine is negative in QIII).

Double angle tangent:

$$\tan 2\theta = \frac{2 \tan \theta}{1 - \tan^2 \theta}$$

$$\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{-3/5}{-4/5} = \frac{3}{4}$$

So:

$$\tan 2\theta = \frac{2 \cdot \frac{3}{4}}{1 - (\frac{3}{4})^2} = \frac{\frac{6}{4}}{1 - \frac{9}{16}} = \frac{\frac{3}{2}}{\frac{7}{16}} = \frac{3}{2} \times \frac{16}{7} = \frac{24}{7}$$

Conclusion: $\boxed{\tan 2\theta = \frac{24}{7}}$

5. Let α satisfy $\tan \alpha = 2$ and α is in the second quadrant. What is the value of $\sin(2\alpha)$?

Solution

- $\tan \alpha = 2$, α in QII ($\sin \alpha > 0$, $\cos \alpha < 0$).
- Recall: $\sin(2\alpha) = 2 \sin \alpha \cos \alpha$.

Let's consider a right triangle with:

- Opposite: 2
- Adjacent: 1
- Hypotenuse: $\sqrt{1^2 + 2^2} = \sqrt{5}$

So:

- $\sin \alpha = \text{positive } \frac{2}{\sqrt{5}}$
- $\cos \alpha = \text{negative } \left(-\frac{1}{\sqrt{5}}\right)$

Therefore:

$$\sin(2\alpha) = 2 \sin \alpha \cos \alpha = 2 \times \frac{2}{\sqrt{5}} \times \left(-\frac{1}{\sqrt{5}}\right) = 2 \times \frac{-2}{5} = -\frac{4}{5}$$

Conclusion: $\sin(2\alpha) = -\frac{4}{5}$

6. An observer is 1 mile from a launch pad. Two rockets are launched at the same time. Rocket A is directly overhead at an angle of elevation that increases from 30° to 60° in 10 seconds. Rocket B's angle of elevation increases from 45° to 75° in the same time. Which rocket had the greater average rate of increase in vertical height, and by how much? (Give an exact or simplified answer.)

Solution

Let the observer be at point O, launch pad at point L. Rockets go straight up from L; O is 1 mile from L.

For both rockets, the **vertical height** at elevation θ is:

$$h = (\text{distance from O to L}) \cdot \tan \theta = 1 \cdot \tan \theta = \tan \theta \text{ miles}$$

Thus, average vertical speed is $\frac{\tan(\theta_2) - \tan(\theta_1)}{\Delta t}$ in miles per second.

Rocket A:

- $\theta_1 = 30^\circ, \theta_2 = 60^\circ$
- $\tan 30^\circ = \frac{1}{\sqrt{3}}$
- $\tan 60^\circ = \sqrt{3}$

So, average rate:

$$r_A = \frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{10} = \frac{\sqrt{3} - \frac{1}{\sqrt{3}}}{10} = \frac{3 - 1}{\sqrt{3} \times 10} = \frac{2}{10\sqrt{3}} = \frac{1}{5\sqrt{3}} \text{ miles/sec}$$

Rocket B:

- $\theta_1 = 45^\circ, \theta_2 = 75^\circ$
- $\tan 45^\circ = 1$
- $\tan 75^\circ = \tan(45^\circ + 30^\circ)$

Recall:

$$\tan(a + b) = \frac{\tan a + \tan b}{1 - \tan a \tan b}$$

So,

$$\tan 75^\circ = \frac{1 + \frac{1}{\sqrt{3}}}{1 - 1 \cdot \frac{1}{\sqrt{3}}} = \frac{\frac{\sqrt{3}+1}{\sqrt{3}}}{1 - \frac{1}{\sqrt{3}}} = \frac{\sqrt{3} + 1}{\sqrt{3}} \cdot \frac{\sqrt{3}}{\sqrt{3} - 1} = \frac{(\sqrt{3} + 1)}{(\sqrt{3} - 1)}$$

So, average rate:

$$\begin{aligned} r_B &= \frac{\tan 75^\circ - \tan 45^\circ}{10} = \frac{\frac{\sqrt{3}+1}{\sqrt{3}-1} - 1}{10} \\ &= \frac{\frac{\sqrt{3}+1-(\sqrt{3}-1)}{\sqrt{3}-1}}{10} = \frac{\frac{2}{\sqrt{3}-1}}{10} = \frac{2}{10(\sqrt{3}-1)} = \frac{1}{5(\sqrt{3}-1)} \end{aligned}$$

Compare $\frac{1}{5\sqrt{3}}$ **and** $\frac{1}{5(\sqrt{3}-1)}$:

Since $\sqrt{3} \approx 1.732$:

- $5\sqrt{3} \approx 8.660 \implies \approx 0.115$
- $5(\sqrt{3} - 1) \approx 3.660 \implies \approx 0.273$

So $\frac{1}{5(\sqrt{3}-1)} > \frac{1}{5\sqrt{3}}$.

Difference:

$$\frac{1}{5(\sqrt{3}-1)} - \frac{1}{5\sqrt{3}} = \frac{1}{5} \left(\frac{1}{\sqrt{3}-1} - \frac{1}{\sqrt{3}} \right) = \frac{1}{5} \left(\frac{\sqrt{3} - (\sqrt{3}-1)}{\sqrt{3}(\sqrt{3}-1)} \right) = \frac{1}{5} \left(\frac{1}{\sqrt{3}(\sqrt{3}-1)} \right)$$

Therefore, **Rocket B had the greater average rate**, and the difference is

1
$5\sqrt{3}(\sqrt{3}-1)$

 miles per second.

Final answer:

Rocket B had the greater average vertical height increase, by $\frac{1}{5\sqrt{3}(\sqrt{3}-1)}$